Integration on Manifolds

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1 Motivation

This lecture completes the lift of analysis from charts to manifolds: after differentiation (tangent vectors, covariant derivative, Lie derivative) we need *integration*. It will be needed to formulate the field equations of general relativity via an action principle (Hilbert's way) rather than by guesswork. As for differentiation, integration needs more structure than a smooth manifold provides, and here the structure is a *volume form*, which in physics is built from a metric (and an orientation).

2 Integration on $\mathbb{R}^d$

A familiar recipe, in two steps:

1. On a *box-shaped domain* $D = [a_1, b_1] \times \cdots \times [a_d, b_d]$ the (Riemann) integral of a function f is the iterated one-dimensional integral, $\int_D f \, \mathrm{d}^d x = \int_{a_1}^{b_1} \cdots \int_{a_d}^{b_d} f(x) \, \mathrm{d}x^d \cdots \mathrm{d}x^1$.

2. For a more general domain $D \subseteq \mathbb{R}^d$, multiply by its *indicator function* χ_D (1 inside, 0 outside) and integrate over all of $\mathbb{R}^d$ (or a box containing D): $\int_D f \, \mathrm{d}^d x := \int_{\mathbb{R}^d} \chi_D \, f \, \mathrm{d}^d x$.

The fundamental tool for practical calculations is the *change of variables*: if $y = \Phi(x)$ is a diffeomorphism from D onto $\Phi(D)$,

$$\int_{\Phi(D)} f(y) \, \mathrm{d}^d y = \int_D f(\Phi(x)) \left| \det \frac{\partial y}{\partial x} \right| \mathrm{d}^d x. \tag{1}$$

Warning. The *absolute value* of the Jacobian determinant appears. Without it the formula is false when $\det < 0$ (the map reverses orientation). Remember this: it is the crux of what follows.

Example 1 (Polar coordinates)**.** $\mathrm{d}^2 x = r \, \mathrm{d}r \, \mathrm{d}\varphi$ because $|\det \partial(x, y)/\partial(r, \varphi)| = r$.

3 First attempt on a manifold, and the problem

Let $(M, \mathcal{O}, \mathcal{A})$ be a d-dimensional smooth manifold, $f : M \to \mathbb{R}$ a function and (U, x) a chart. The naive definition

$$\int_U f \text{ “}=\text{”} \int_{x(U)} (f \circ x^{-1}) \, \mathrm{d}^d x$$

depends on the chart: if (U, y) is another chart on the same domain, then by (1) the two results differ by the Jacobian factor, so this is *not* a well defined integral of f.

We therefore look for an extra object: a quantity with a component function $\omega_{(x)}$ in every chart, such that

$$\int_{x(U)} (f \circ x^{-1}) \, \omega_{(x)} \, \mathrm{d}^d x$$

is chart independent. Using (1), with $\hat{f}_x = f \circ x^{-1}$ and $\hat{f}_y = f \circ y^{-1}$,

$$\int_{y(U)} \hat{f}_y \, \omega_{(y)} \, \mathrm{d}^d y = \int_{x(U)} \hat{f}_x \, \omega_{(y)} \, \Big| \det \frac{\partial y}{\partial x} \Big| \, \mathrm{d}^d x,$$

so we need

$$\boxed{\omega_{(x)} = \omega_{(y)} \, \Big| \det \frac{\partial y}{\partial x} \Big|} \qquad \Longleftrightarrow \qquad \omega_{(y)} = \Big| \det \frac{\partial x}{\partial y} \Big| \, \omega_{(x)}. \tag{2}$$

Such a "density" is not a function (functions do not transform at all), but it *does* resemble something we know: the components of the metric.

4 The volume form

Under a change of chart, $g_{(y)} = \big(\frac{\partial x}{\partial y}\big)^T g_{(x)} \big(\frac{\partial x}{\partial y}\big)$, hence

$$\det g_{(y)} = \Big(\det \frac{\partial x}{\partial y} \Big)^2 \det g_{(x)}, \qquad \sqrt{|\det g_{(y)}|} = \Big| \det \frac{\partial x}{\partial y} \Big| \sqrt{|\det g_{(x)}|},$$

which is exactly the required law (2) for $\omega_{(x)} := \sqrt{|\det g_{(x)}|}$.

4.1 Levi-Civita symbol and orientation

The d-form that does this is built with the *Levi-Civita symbol* $\varepsilon_{i_1 \dots i_d} \in \{0, \pm 1\}$: it is $+1$ for $(i_1, \dots, i_d)$ an even permutation of $(1, \dots, d)$, -1 for an odd one, and 0 otherwise. It is a symbol defined in each chart (not a tensor); totally antisymmetric.

A d-form has only one independent component, and it transforms with the determinant itself, $\det(\partial x / \partial y)$, *not* with its absolute value. To make this agree with (2) one needs the determinant to be positive:

Definition 1 (Oriented atlas)**.** An atlas $\mathcal{A}$ is *oriented* if for any two charts $(U, x), (V, y) \in \mathcal{A}$ the transition map has positive Jacobian determinant, $\det \frac{\partial y}{\partial x} > 0$, on $U \cap V$. A manifold is *orientable* if it admits an oriented atlas (e.g. spheres are, the Möbius strip is not). Choosing one, one speaks of an *oriented manifold* and restricts to the oriented atlas.

Definition 2 (Volume form)**.** Let (M, g) be an oriented metric manifold. The *volume form* is the d-form ω whose components in every chart of the oriented atlas are

$$\omega^{(x)}_{i_1 \dots i_d} = \sqrt{|\det g_{(x)}|} \, \varepsilon_{i_1 \dots i_d}, \qquad \text{i.e.} \quad \omega = \sqrt{|\det g_{(x)}|} \, \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^d.$$

Theorem 1. *These component functions transform as the components of a d-form between charts of an oriented atlas, so ω is a well defined, chart-independent object.*

Proof. A d-form's component transforms with $\det(\partial x / \partial y)$, while the formula above transforms with $|\det(\partial x / \partial y)|$; the two agree when the Jacobian is positive, which holds in an oriented atlas. □

Remark. All the information needed to *measure volumes* lies in this one object. A metric is not the only way to get one (a volume form or density can also be specified directly), but a metric together with an orientation always determines it. The additional structure is required: on a bare smooth manifold there is no way to say how large a region is.

Example 2. On $\mathbb{R}^2$ with the Euclidean metric in polar coordinates, $g = \mathrm{diag}(1, r^2)$, so $\sqrt{|\det g|} = r$ and $\omega = r \, \mathrm{d}r \wedge \mathrm{d}\varphi$. On the round S^2, $g = \mathrm{diag}(1, \sin^2\theta)$, so $\omega = \sin\theta \, \mathrm{d}\theta \wedge \mathrm{d}\varphi$, and the total area is $\int_0^{2\pi} \int_0^{\pi} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\varphi = 4\pi$.

5 The integral over a manifold

5.1 Within a single chart

Definition 3. Let (U, x) be a chart of an oriented atlas and f a function on M with support in U (or integrable there). Define

$$\int_U f\,\omega := \int_{x(U)} (f \circ x^{-1})\,\sqrt{|\det g_{(x)}|}\,\mathrm{d}^d x.$$

By the computation of Section 3 this is independent of the chart used.

5.2 Over all of M: partition of unity

If M is not covered by one chart, we cannot simply add up the chart integrals: overlaps would be *counted twice*. The fix is a *partition of unity*: smooth functions $\rho_\alpha : M \to [0,1]$, one for each chart (U_α, x_α) of the oriented atlas, such that

(i) $\operatorname{supp}\rho_\alpha \subseteq U_\alpha$ (together with local finiteness),

(ii) $\sum_\alpha \rho_\alpha(p) = 1$ for every $p \in M$.

Condition (ii) says: at every point, the contributions from all charts sum to one, so nothing is double counted. Then

Definition 4 (Integral on a manifold)**.** For a function f (with compact support, or otherwise integrable),

$$\int_M f\,\omega := \sum_\alpha \int_{U_\alpha} \rho_\alpha f\,\omega = \sum_\alpha \int_{x_\alpha(U_\alpha)} \big((\rho_\alpha f)\circ x_\alpha^{-1}\big)\sqrt{|\det g_{(x_\alpha)}|}\,\mathrm{d}^d x_\alpha.$$

One shows that the result does not depend on the choice of oriented atlas or of the partition of unity (not worked out here). Constructing partitions of unity is not quite easy, but they exist on the manifolds we consider. In practice one often uses a single chart that covers all but a set of measure zero (e.g. the (θ, φ) chart on S^2), which suffices for the calculation. The integral over a subdomain $D \subseteq M$ is defined with the indicator function, as in $\mathbb{R}^d$.

6 Application: the Einstein–Hilbert action

With integration in hand, the field equations of Einstein's theory can be obtained from an action, as done by Hilbert. For a Lorentzian metric g on a four-dimensional (oriented) manifold, the *Einstein–Hilbert action* is

$$S_{\mathrm{EH}}[g] = \int_M R\,\omega = \int_M R\,\sqrt{|\det g|}\,\mathrm{d}^4 x,$$

where R is the Ricci scalar of the Levi-Civita connection (up to constants such as $1/16\pi G$). The factor $\sqrt{|\det g|}$ is exactly the density that makes this integral chart independent; without it the integrand R alone would not define an invariant integral. The field equations follow by varying S with respect to the metric.

Summary

- On $\mathbb{R}^d$: box domains, indicator functions for general domains, and the change-of-variables formula with $|\det \partial y/\partial x|$.
- The naive chart integral of a function is chart dependent. One needs a density $\omega_{(x)}$ with $\omega_{(y)} = |\det \frac{\partial x}{\partial y}|\omega_{(x)}$.
- The metric supplies one: $\omega_{(x)} = \sqrt{|\det g_{(x)}|}$. On an *oriented* atlas this is the component of the volume form $\omega = \sqrt{|\det g|}\,\varepsilon_{i_1\ldots i_d}$, a genuine d-form.
- $\int_U f\,\omega = \int_{x(U)} (f \circ x^{-1})\sqrt{|\det g|}\,\mathrm{d}^d x$ within a chart; globally one uses a partition of unity to avoid double counting in overlaps.
- Application: the Einstein–Hilbert action $\int_M R\sqrt{|\det g|}\,\mathrm{d}^4 x$.